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TITLE PAGE

- **Problem Statement ID –** SIH25149
- **Problem Statement Title-** Loan Utilization Tracking via Mobile.
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID -** 72262
- **Team Name -** Technical Terminators

Technical Terminators

NYAY SAHAYAK – HYBRID AI POWERED LOAN UTILISATION TRACKING SYSTEM

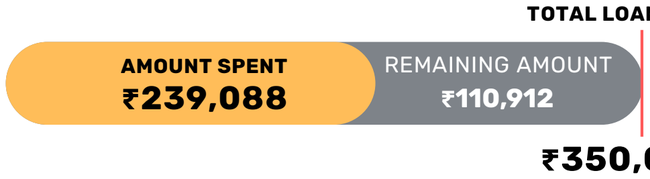
-Ensuring Transparent, Tamper-Proof Loan Utilization for MoSJE Beneficiaries

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Proposed Solution

1. **Beneficiary App** – Lets users **upload** geo-tagged, time-stamped photos/videos of fund usage even in **offline mode**.
2. **AI Verification Engine** – Detects tampered or duplicate submissions automatically, reducing officer workload.
3. **Stage-Wise Utilisation Tracker** – **Visually** shows loan amount used at **each stage**, promoting accountability.
4. **Parivahan API** - **Automates** the verification of commercial vehicles using **RC Book.**
5. **Officer Dashboard** – Enables MoSJE officers to **remotely review**, **approve, and monitor utilisation analytics.**
6. **Secure Cloud Sync** – **Encrypts** and syncs proofs automatically, ensuring **tamper-proof,** auditable verification.

LOAN AMOUNT UTILISATION TRACKING SYSTEM-PROGRESS BAR



USE CASE: FOR STAGE WISE LOAN AMOUNT UTILISATION TRACKING FOR BANK OFFICERS AND BENEFICIERIES

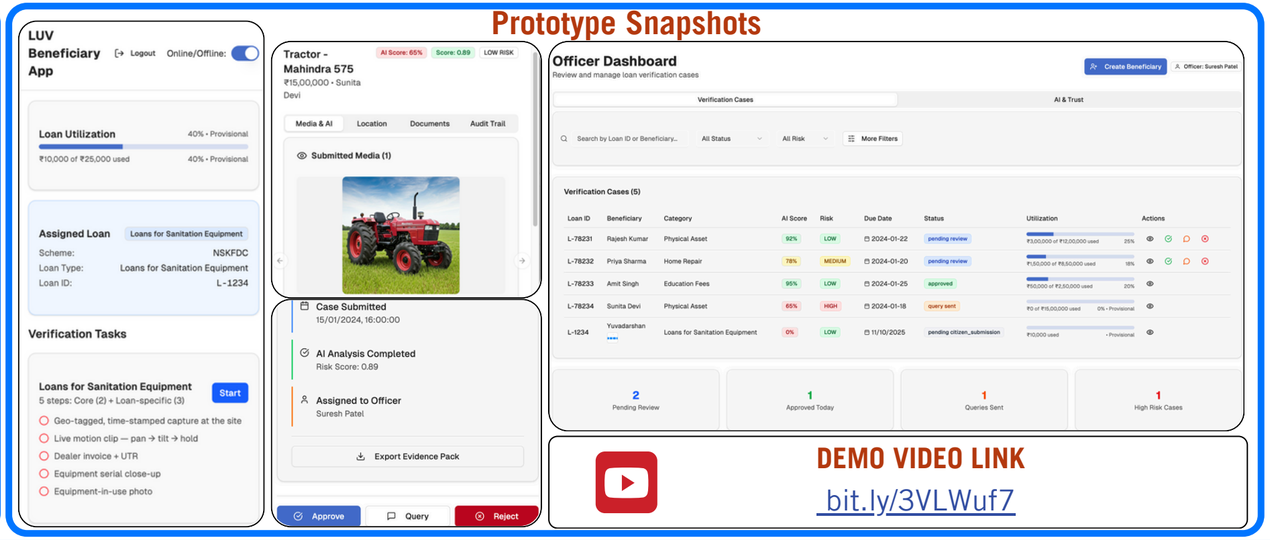
TYPES OF LOAN SHEMES - MoSJE



Innovation & Uniqueness

- **ZK Geofence Proof** — Confirms correct location **without sharing GPS** details.
- **Live Video + IMU Lock** — Verifies real-time, on-site evidence authenticity of live videos.
- **Semantic Progress AI** — Detects actual **progress of utilization stages**.
- **Predictive Risk Scoring** — **Flags suspicious** or inconsistent submissions **early**.
- **Private Set Intersection** — **Prevents duplicate** or reused invoice fraud.
- **Merkle Evidence Pack** (**Blockchain**) — **Ensures tamper-proof** and auditable verification.
- **Explainable AI Assistance —** for officers with clear **fraud reasoning for accept/reject.**

Prototype Snapshots



DEMO VIDEO LINK

bit.ly/3VLWuf7

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TECHNICAL OVERVIEW

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BANK LOGIN

Username

Password

Forgot Password?

Login

1

Create Loan Beneficiary

Create Beneficiary Loan Verification

Loan ID

Enter loan ID

Phone Number

+91 10-digit number

Name (Optional)

Beneficiary name

Category of Loan

Select loan category

Loan Agreement (PDF)

Choose File no file selected

Cancel Create

Officer enters
-Loan ID,
-Phone No.
-Category
-Loan Documents

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Officer Dashboard

Verification Cases (5)

Loan ID	Beneficiary	Category	AI Score	Risk	Due Date	Status
L-78231	Rajesh Kumar	Physical Asset	92%	LOW	+ 2024-01-22	pending review
L-78232	Priya Sharma	Home Repair	78%	MEDIUM	+ 2024-01-20	pending review
L-78233	Amit Singh	Education Fees	95%	LOW	+ 2024-01-25	approved
L-78234	Sunita Devi	Physical Asset	65%	HIGH	+ 2024-01-18	query sent
1234567890	asn	Vehicle	0%	LOW	+ 10/2/2025	pending citizen

12

APPROVED

REJECTED

Decision updates, citizen notified instantly

10

2

MAIN SERVER

3

CITIZEN LOGIN

Login

4

UPLOAD ASSETS
Geo-tagged photos, videos, invoices, documents, evidence

5

AI Pre-screening

AI Pre-Screening
Detect Invalid/Blurry Photos

6

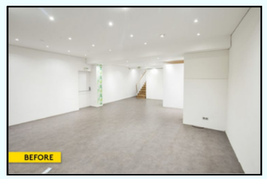
LOCAL STORAGE
Store and access data locally on the device without internet.

7

9

8

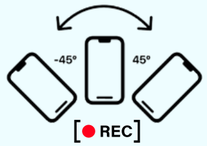
AI ENGINE





SEMANTIC PROGRESS VERIFIER
Measure % of work done of same-spot photos of Shops

GEO-NET
Verifies capture occurred inside the allowed site (GPS)



Live Motion+Pathlock
Verifies authenticity with live motion-sensor guided actions to prevent replay fraud.

FRAUD DETECTION
Anomalies and duplicates flagged early

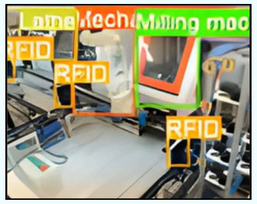
FRAUD ALERT

De-Duplication
Removes duplicate loan applications using ID, docs, and asset checks.

CLASSIFIED

JPG

CNN MEDIA CLASSIFIER
Photo–asset match against loan requisition



TECH STACK

Ai Algorithm
- LSTMs/GRUs
- Transformer-based NLP (BERT/DistilBERT)
- Autoencoders and GNNs

Frontend
- React Native
- React
- Vite
- Map/Geo-tagging APIs

Backend
- Node.js
- Python
- Microservices Architecture.
- Document Verification Service

DataBase
- PostgreSQL / MySQL
- MongoDB
- Object Storage S3
- GCP

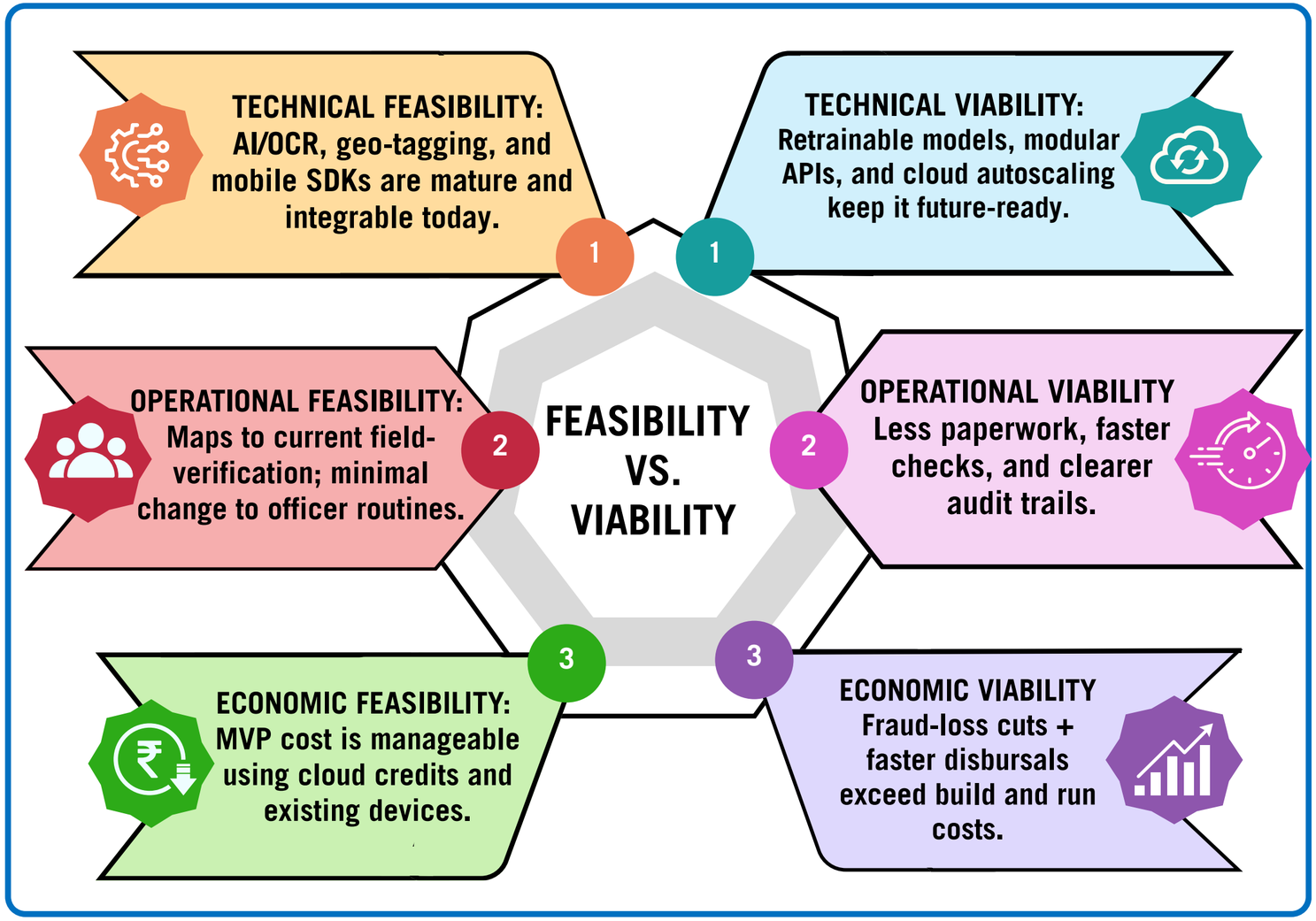
Infra
- Docker
- Kubernetes
- CI/CD

APIs
- RESTful API
- GraphQL
- WebSockets

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FEASIBILITY AND VIABILITY

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- Fear of Mistakes
- Patchy Networks
- Low-End Device Struggle
- Privacy Hesitation
- Review Load Spikes

CHALLENGES

- AI Sahayak & Training
- Offline & Sync feature
- Pilot First Approach
- Privacy by Design
- Anti-Fraud & Ops

MITIGATIONS

Implementation Strategy

- **Phase 1 (Pilot):** Deploy in 1 state under MoSJE for 2–3 loan schemes.
- **Phase 2 (Optimization):** Integrate AI risk scoring & anomaly detection.
- **Phase 3 (Nationwide Rollout):** Extend to all state agencies with retrainable models.

Goal: 6–9 months full deployment cycle.

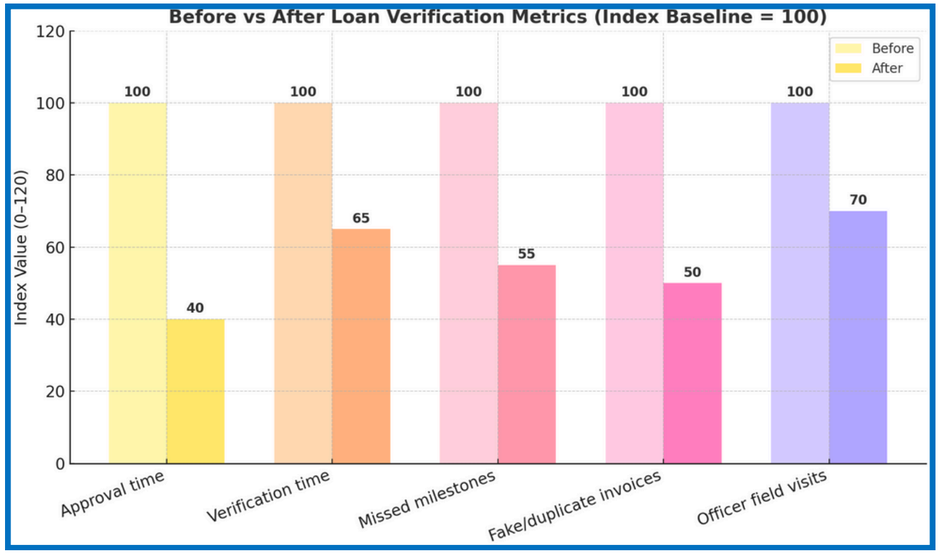
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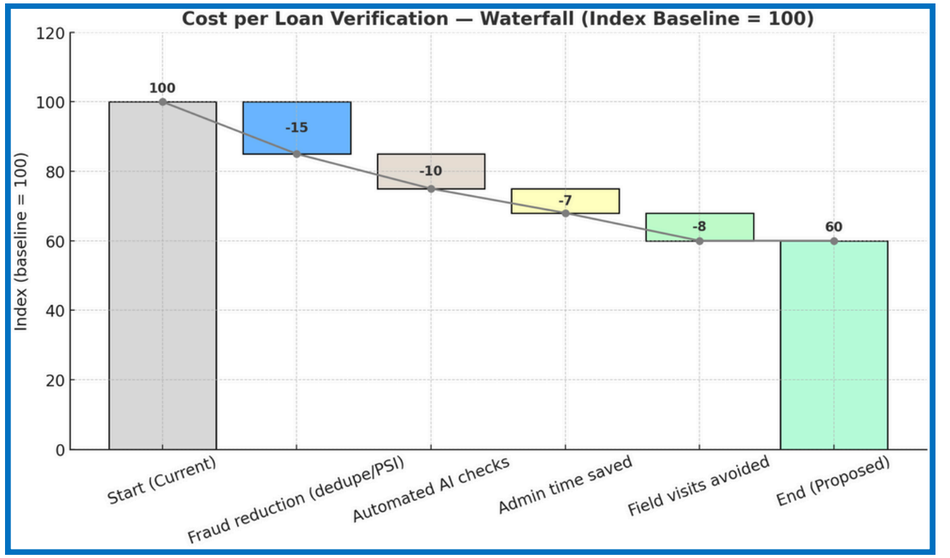
IMPACT AND BENEFITS

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IMPACTS	BENEFITS
FRAUD REDUCTION & TRUST - Misuse and diversion ↓ 40–60% [5] - Faster approvals (TAT ↓ 25–35%) [1] build citizen confidence.	**SAFER CREDIT ECOSYSTEM** - Funds used as intended [5]; - Defaults/fake claims ↓ 20–35% [5].
OPERATIONAL EFFICIENCY & COST - Manual checks ↓ 30–45% [3] - Field visits ↓ 20–30% [3]	**ECONOMIC EFFICIENCY** - Admin work ↓ 20–30% [2] Disbursement cycle time ↓ 25–40% [1]
FINANCIAL INCLUSION - Trust Score ↑ 0.5–0.8 tiers [2] - Repeat loans ↑ 20–30% [1]	**ACCESS & UPTAKE** - Lower rates / instant top-ups via Trust Score [2] - Participation ↑ 30–50% [1]
EFFICIENT GOVERNANCE - SLA ↑ 25–35% [4] - Anomaly flagging ↑ 40–55% [5] - Merkle logs	**TRANSPARENCY & ACCOUNTABILITY** - Tamper-proof audit [5] - Disputes ↓ 40–50% [4]

Refs: [1] KPMG India | [2] World Bank (2023) | [3] RBI (2025) | [4] CKYC (official) | [5] SWIFT × MonetaGo





Technical Terminators

RESEARCH AND REFERENCES

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